Dr. JAYANTA PAUL

Resume

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Education

2025 **PhD, Computer Science & Engineering**, *Indian Institute of Engineering Science and Technology (IIEST)*, Shibpur(*Formerly* Bengal Engineering and Science University, Shibpur), *Specialization on:* Machine Learning, Deep Learning, Computer Vision, Natural language processing.
Thesis Title: Deep Learning-Driven Framework for Multimodal Social Media Content Analysis; Thesis Defense Completed on: 27.11.2025, Date of Award: 19.12.2025

2020 **Master of Technology, Computer Science & Engineering**, *Indian Institute of Engineering Science & Technology*, Shibpur(*Formerly* Bengal Engineering and Science University, Shibpur).

2017 **Bachelor of Technology, Computer Science & Engineering**, *Maulana Abul Kalam Azad University of Technology, West Bengal(MAKAUT)* , (*Formerly* West Bengal University of Technology).

Work Experience

July 2026 – Present **Institute Post-Doctoral Fellow**, *Department of Artificial Intelligence, Indian Institute of Technology, Kharagpur, West Bengal, India - 721302.*.

April 2026 – July 2026 **CHANAKYA Postdoctoral Fellow**, *AI4ICPS I Hub Foundation and Department of Artificial Intelligence, Indian Institute of Technology, Kharagpur, West Bengal, India - 721302.*.

August 2025 – April 2026 **Assistant Professor**, *Department of Computer Application and Science, Institute of Engineering and Management (IEM), Kolkata, 700091*, Ashram Building, Sector-V, Salt Lake Electronics Complex.

Publications

Journal Articles

2026 Abhijit Mitra, Souvik Mondal, Anubhav Bhattacharya, **Jayanta Paul**, and Jaya Sil., Mask-guided rationales meet LLMs: Hybrid encoder–LLM question answering framework for grounded, faithfulness-regularized explainable hate speech detection., In ***Applied Soft Computing, volume 200, article 115373. Elsevier***.

2026 Abhijit Mitra, Rajdeep Majumdar, Rivujit Das, Arghyadeep Saha, **Jayanta Paul**, Souvik Mondal, and Jaya Sil., Breaking Bias: A Context-Aware Multimodal Framework for Detecting and Neutralizing Ideological Bias in News., In ***ACM Transactions on Knowledge Discovery from Data, volume 20, issue 3, article 38, pages 1–69. ACM***.

2025 **Jayanta Paul**, Anuska Roy, Abhijit Mitra, and Jaya Sil., HyV-Summ: Social media video summarization on custom dataset using hybrid techniques., In ***Neurocomputing, volume 614, article 128852. Elsevier***.

2025 Abhijit Mitra, **Jayanta Paul**, Tanis Ahamed, Sagar Basak, and Jaya Sil., From Words to Visuals: Bridging Text and Visual Insights Using MetA-MARC Framework for Enhanced Scholarly Article Categorization., In ***Knowledge-Based Systems, volume 324, article 113896. Elsevier***.

2025 **Jayanta Paul**, Siddhartha Mallick, Abhijit Mitra, Anuska Roy, and Jaya Sil., Multi-Modal Twitter Data Analysis for Identifying Offensive Posts Using a Deep Cross-Attention-Based Transformer Framework., In ***ACM Transactions on Knowledge Discovery from Data, volume 19, issue 3, article 66, pages 1–30. ACM***.

2025 **Jayanta Paul**, Abhijit Mitra, Divyanshu Kulhadiya, Tejas Pawar, Angshuman Roy, and Jaya Sil., A Comprehensive Approach to Real-Time Attendance Systems: Integrating Face Recognition and Emotion Detection with Web Technologies., In ***Procedia Computer Science, volume 258, pages 3436–3446. Elsevier***.

2024 **Jayanta Paul**, Ahel Das Chatterjee, Devtanu Misra, Sounak Majumder, Sayak Rana, Malay Gain, Anish De, Siddhartha Mallick, and Jaya Sil., A Survey and Comparative Study on Negative Sentiment Analysis in Social Media Data., In ***Multimedia Tools and Applications, volume 83, issue 30, pages 75243–75292. Springer***.

2023 Siddhartha Mallick, **Jayanta Paul**, and Jaya Sil., Response Fusion Attention U-ConvNext for Accurate Segmentation of Optic Disc and Optic Cup., In ***Neurocomputing, volume 559, article 126798. Elsevier***.

2023 Rajat Subhra Bhowmick, Rahul Indra, Isha Ganguli, **Jayanta Paul**, and Jaya Sil., Breaking CAPTCHA System with Minimal Exertion through Deep Learning: Real-Time Risk Assessment on Indian Government Websites., In ***Digital Threats: Research and Practice, volume 4, issue 2, article 28, pages 1–24. ACM***.

2023 Rajat Subhra Bhowmick, Isha Ganguli, Ananya Paul, **Jayanta Paul**, and Jaya Sil., Improving Indic Code-Mixed to Monolingual Translation Using Mixed Script Augmentation, Generation & Transfer Learning., In ***ACM Transactions on Asian and Low-Resource Language Information Processing. ACM***.

2022 Rajat Subhra Bhowmick, Isha Ganguli, **Jayanta Paul**, and Jaya Sil., A Multimodal Deep Framework for Derogatory Social Media Post Identification of a Recognized Person., In ***ACM Transactions on Asian and Low-Resource Language Information Processing, volume 21, issue 1, article 2, pages 1–19. ACM***.

Manuscripts Under Review / Communicated

2025 **Jayanta Paul**, Abhijit Mitra, and Jaya Sil., IndiToon MemeNet: A Deep Learning Framework for Unethical Content Detection and Person Recognition in Indian Cartoon Memes., *Information Fusion, Elsevier* – Communicated.

2025 Siddhartha Mallick, Aayushman Ghosh, **Jayanta Paul**, and Jaya Sil., CATFA-Net: A Trans-Convolutional Approach for Accurate Medical Image Segmentation., *Medical Image Analysis, Elsevier* – Under revision.

Conference and Book Chapter

2025 **Jayanta Paul**, Sanju De, Angshuman Roy, Arkapravo Das, and Abhijit Mitra, Deep Learning-Powered IoT System for Real-Time Environmental Health Risk Monitoring and Proactive Alerts, In ***7th International Symposium on Devices, Circuits and Systems May, 2025*** .

2024 **Jayanta Paul**, *Abhijit Mitra, Somak Sanyal, and Jaya Sil.*, Socialfaceemonet: A deep architecture for social media face recognition and emotion detection using customized datasets., In ***International Conference on Pattern Recognition, pages 309–327. Springer***.

2024 Abhijit Mitra, Rivujit Das, **Jayanta Paul**, Rajdeep Majumder, Arghyadeep Saha, Nandita Sengupta, and Jaya Sil., A deep learning framework for classifying and mitigating bias in news reporting., In ***Proceedings of the Future Technologies Conference, pages 424–443. Springer***.

2023 Siddhartha Mallick, Nitu Saha, **Jayanta Paul**, Isha Ganguli, Shantonu Debnath, and Jaya Sil., An efficient deep learning framework for glaucoma diagnosis using convolution mixed transformer network. , In ***TENCON 2023-2023 IEEE Region 10 Conference (TENCON), pages 1111–1116. IEEE***.

2022 **Jayanta Paul**, Anuska Roy, Siddhartha Mallick, and Jaya Sil. , A comparative study of deep learning-based face recognition and emotion detection techniques using social media customized cartoon post. , In ***In International Conference on Computational Intelligence in Pattern Recognition, pages 401–411***.

2022 **Jayanta Paul**,Rajat Subhra Bhowmick, and Jaya Sil. , Low-computation iot system framework for face recognition using deep learning algorithm. , In ***International Conference on Computational Intelligence in Pattern Recognition, pages 24–35. Springer***.

2022 Siddhartha Mallick,**Jayanta Paul**,Nandita Sengupta, and Jaya Sil. , Study of different transformer based networks for glaucoma detection. , In ***TENCON 2022-2022 IEEE Region 10 Conference (TENCON), pages 1–6. IEEE***.

2021 **Jayanta Paul**, Rajat Subhra Bhowmick, Baisakhi Das, and Biplab K Sikdar. , A smart home security system in low computing iot environment. , In ***IEEE 17th India council international conference (INDICON), pages 1–7. IEEE***.

2020 **Jayanta Paul**,Rajat S Bhowmick, Riom Sen, Dwaipayan Ray, Suman S Manjhi, Soumya Sen, and Biplab K Sikdar. , Evaluation of face recognition schemes for low-computation iot system design, In ***24th International Symposium on VLSI Design and Test (VDAT), pages 1–6.IEEE***.

Fellowships & Awards

2026 – 2027 ***Chanakya Fellowship*** CHANAKYA Postdoctoral Fellow at the Department Artificial Intelligence and AI4ICPS Innovation Hub, IIT Kharagpur.

2020 – 2025 ***Institute Fellowship*** as a Ph.D. research scholar at Indian Institute of Engineering Science and Technology (IIEST), Shibpur.

2017 Achieved 1st position making quadcopter (robotics championship zonal round) under ARK Techno solutions and robokart.com (IIT Madras).

Academic Achievements & Recognitions

2026 Invited to take the lecture on ***"Mobile & Digital Trace Data Using Mobility Patterns for Urban Planning"*** during the Residential Training Programme on Big Data, AI, and ML using Python for the ISS Probationers funded by the **Ministry of Statistics and Programme Implementation (MoSPI), Govt of India and organized by the IIT Kharagpur AI4ICPS I Hub Foundation at IIT Kharagpur** on 31st May, 2026.

2023 Invited for a talk in ***"The Innovative Power of Machine Learning in Image Processing"***, during 6th-10th November, 2023 at **Faculty Development Programme** Organization- MCKV Institute of Engineering, Lilua, Howrah.

2017 Qualified GATE Examination

Technical skills

Programming Languages: Python, PyTorch, TensorFlow, C

Subject: Machine learning, Deep learning, Computer Vision.

Position of Responsibility

2014-2017 **Class representative in CSE Depertment**, Global Institute Of Management And Technology.

2023-2024 **Year representative**, *Ph.D.*, CST Department, Indian Institute of Engineering Science and Technology (IIEST), Shibpur.

2023-2024 **Treasurer**, *Academic Society of Computer Engineers*, CST Department, Indian Institute of Engineering Science and Technology (IIEST), Shibpur.

March 22-24, 2023 **Organizer, Revelation**, *The Departmental Fest of CST* , Indian Institute of Engineering Science and Technology (IIEST), Shibpur.

April 20-23, 2023 **Organizer, Rebeca**, *The 82nd edition of the REBECA 2023 cultural festival at*, Indian Institute of Engineering Science and Technology (IIEST), Shibpur.

April 4-6, 2024 **Committee Convenors, Revelation**, *The Departmental Fest of CST* , Indian Institute of Engineering Science and Technology (IIEST), Shibpur.

March 21-23, 2025 **ML Event Lead, Revelation**, *Academic Society of Computer Engineers*, CST Department, Indian Institute of Engineering Science and Technology (IIEST), Shibpur.

Laboratory Assistantship

2022-2024 : **CS4271: Artificial Intelligence Lab**, Indian Institute of Engineering Science and Technology (IIEST).

2021-2023 : **CS4172: Machine Learning Laboratory**, Indian Institute of Engineering Science and Technology (IIEST).

2019-2024 : **CS1271: Introduction to Computing**, Indian Institute of Engineering Science and Technology (IIEST).

2022-2024 : **CS3273: Software Engineering Lab**, Indian Institute of Engineering Science and Technology (IIEST).

2021-2024 : **CS3183: Soft Computing Lab**, Indian Institute of Engineering Science and Technology (IIEST).

References

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Personal Information

Father : Mr. Ranjan Kumar Paul

Gender : Male

Languages : Bengali, Hindi and English

Hobby : Playing cricket, playing guitar, teaching, editing videos

DOB : 10.02.1995